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Performance and Salary Estimates in Professional Baseball*
: Evidence from Korean Free Agent Players

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This paper investigates performance and salary estimates for Korean free agent(FA) players in professional baseball league. We estimate marginal revenue products of free agent Korean professional baseball players using multiple regression analysis. We first construct a panel data of Korean Professional Baseball League official records from 2000 to 2018 to formulate two equations: team-winning function and team-revenue function which is a modified version of Scully(1974). The estimates for marginal revenue product(MRP) of a player in Korean Baseball Organization(KBO) are derived from team-level performance, team winning, and team revenue. For robustness purposes, recent performance measures of sabermetrics are used. We empirically evaluate MRP of free agent players. MRP estimates of Lee Dae-ho and Yang Hyun-jong are compared with their actual salaries. The results suggest that both players are underpaid in pre-FA period and are overpaid in post-FA period.

※ Key Words : Korean Baseball Organization, Multiple regression analysis, Marginal revenue products, Salary estimation, Sabermetrics

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Introduction

Among many professional sports, baseball is an interesting laboratory for labor economists because it is one of the easiest sports to express in numbers. Every plays and moves are being recorded and cumulated to form a collective data. Bill James is viewed as pioneer of advanced baseball statistics, Sabermetrics. Sabermetrics is the field of analyzing records of professional baseball that allows baseball more objective and predictable. It allows team directors and managers to build up strategies, evaluate a player, and negotiate player's salary. Amid thousands of fans and spotlights, some of professional baseball players earn billions of salaries while some are given minimum salaries. In fact, Among 513 registered KBO professional players, Lee Dae-ho's annual salary is 2.5 billion won , but 113 players are receiving the minimum annual salary of 27 million won. There are not many professional sports players compare to other jobs. They are physically outstanding, and also dedicated to excel their unique skills sets. It is inappropriate to measure an annual salary based on the number of hours worked like a normal worker. Unlike ordinary companies that let one promotes based on timely manner, the record of performance is the most important criteria for determining annual salary for professional baseball players (Lee and Kang, 2001). However, there are constant controversies with overrated salaries and lack of performance. Plausible explanation is required for Lee Dae-ho earning 100 times more than a minimum annual salary player.

There are other significant factors that affect the salary other than performance. One of them is the salary negotiation. The KBO has a term called reserve clause that gives the franchised team the exclusive right to renew and terminate the contract. Professional players in KBO have to be registered and play for nine years under the reserve clause to become a free agent player. Before becoming a free agent player, one cannot negotiate the salary. Under the reserve clause, KBO players are known to be severely underpaid (Han and Jung, 2012). There are also studies that salary disparity and perception of unfairness may lead to lower morale which eventually connects to inferior performance (DeBrock, Hendricks, and Koenker, 2004).

Evaluation of the relationship between player's performance and salary is important to both management and policy maker in sports industry. The economics of professional team sports and significance of sports economics in KBO, however, are not as developed as MLB. It is shorter and smaller in terms of history and size of the market. Extensive data and studies have to be cumulated to estimate reliable connection between the statistical and financial value. A further study on performance and salary evaluation is necessary to meet the growing popularity in KBO.

A literature review is presented in the remainder of Section 1. The History of Korean professional baseball league is also outlined. Section 2 presents the empirical specification of the model. Section 3 describes the data. Section 4 presents estimation results that are based on three specifications for the team winning percentage model, and results for the team revenue model are also discussed. Section 5 concludes the paper.

Literature Review

Research on player's performance and salary evaluation has been steadily and diversely conducted for MLB. Scully (1974), a pioneer of economy of Major League baseball, established a model to estimate marginal revenue of products (MRP) of players and to prove absurdity of reserve clauses. Scully (1974) estimated the team percent win function PCTWIN which measures the influence of batting and pitching power on the team win rate. The team revenue function was estimated with the variables of PCTWIN, tickets sold, broadcasting rights, and population. Then the MRP of a player could be estimated from the two functions. He concluded that professional baseball players in 1968 to 1969 were exploited under reserve clause and monopsony power.

Sommer and Quinton (1982) used the same method to estimate the MRP of players that Scully (1974) used in his paper. But he focused on the impact of Free Agent and salaries of "first family" of free agents in 1976 that became free from reserve clause and monopsonistic characteristic of the market. Free Agent players were believed to be overpaid, but found to be fair. Free Agent players earned their salaries equal to the marginal revenue product estimated. Sommer and Quinton (1982) also found that average ballplayers were still severely underpaid under the reserve clause. Annala and Winfree (2011) and Kahn (2010) also provide insight on player's performance and compensation in MLB with similar opinions.

Lee and Kang (2001) investigated the relationship between the pitcher's performance and the arrangement of salary in KBO. The result of study illustrates that pitcher's annual salary is affected by performance indicators such as Earned Run Average, Strikeouts, and Wins and Losses. It suggests that a KBO player's salary is actually influenced by both individual and team related factors.

Kim (2002) had similar empirical analysis on performance and salary of KBO pitchers from 1999 to 2001. Setting annual salaries as dependent variable and ERA, the number of appearances, wins, losses, saves, the innings pitched, hits, hits by pitch, strikeouts, errors as independent variables, Kim (2002) used principal component regression analysis. The analysis found that the annual salary of pitchers is not determined by the player's actual performance of the season, but rather broad concept of contribution to the team winning. The variables that had the greatest impact on the salary were the number of wins and saves for the year. Average annual salary increased every year, but the number of wins and saves decreased.

Hong et al. (2015) conducted discrimination and classification analysis on 65 KBO batters who signed contracts in the FA market from 2006 to 2012 to evaluate salary and performance. They used variables such as batting average, homerun, steal, walk, and age as variables for average annual salary of FA contracts. They also found that the players who signed FA contracts after 2011 had better salary negotiation despite the relatively poor performance.

Lee (2007) and Lee (2010) had empirical analysis on determinants of team winning and the role of management system. Lee (2007) tried to apply Auckland As's Moneyball management method on KBO using the panel data model and regression analysis. It focused on the record of on-base and slugging percentage, and autonomous management system. As the result, it was estimated that the importance of the on-base percentage was more than three times higher than the slugging percentage which is very similar to Auckland team's analysis. And it was also possible to infer that autonomous baseball produced more effective results on average.

Lee (2010) wrote a paper regarding the influence of so called Smallball which refers to “a manager’s active intervention or strategy implementation in baseball.” Smallball was a common system of baseball management in KBO and played more frequently than MLB. Lee (2010) presented the panel data analysis of a stochastic production frontier model to validate whether the strategy of small ball is actually effective for winning. The result indicated that stealing attempts were positive, but use of bunting and pinch hitters were found to be negative. Furthermore, the overall effects on run production were negative. He concludes that small ball reduces the efficiency of offensive production.

In this study, we compared the actual and estimated salaries of free agent KBO players who are free from reserve clause. We collected data of Korean Professional Baseball Leagues from 2000 to 2018. To estimate marginal revenue products (MRP), we applied a modified version of Scully (1974)’s model. Evaluations of the team revenue model using KBO data have received less attention. In this context, this study contributes to the literature by examining the KBO-specific institutional background and team-specific intensity of fan interest. We estimated how much influence the players' performance indicators have on the winning rate, and how much influence the winning rate has on the team revenue. The coefficients estimated from the two models indicate player’s contribution toward winning and revenue and allow us to acquire the MRP of hitters and pitchers. As a result, we can evaluate salary of free agents players.

History of Korean Professional Baseball League

Korean Professional Baseball League was found in 1982 with six teams. Each team has their sponsored company and represent city. There were modifications and expansion in the league. There are total of 10 franchised teams and they are KIA, Lotte, LG, Doosan, Samsung, Hanwha, SK, Kiwoom, NC, and KT. The league has been rapid growing in the size and quality. Table 1 shows number of games and attendance increases. 532 games with total of 2,507,549 attendances in 2000 have become 720 games with 8,073,742 attendances in 2018. The number of attendances has tripled since.

10 teams are compete against each other in regular season, and only the top five teams are moving on to post season. Post season known also as autumn baseball is four steps of one versus one competition that only a winner can move up. The match-up between fifth and fourth place of a regular season is called the Wild Card. The winner of Wild Card proceeds to the Semi-Playoff matching against the third place. The next is the Playoff, and finally the Korean Series. KIA Tigers won the most Korean Series and took the title 10 times. Samsung Lions follows with 8 championship titles, Doosan Bears with 5 times and SK Wyverns with 3 times. Both Lotte Giants and LG Twins won twice. Hanwha Eagles won only once in 1999. The relatively new team, Kiwoom Heroes, NC Dinos, and KT Wiz have not won the title yet.

The reserve clause, the exclusive rights for sponsored company to trade and sell players at their will, has been applied since the beginning of the KBO. Draft system with regional priority existed since 1983. These practices restricted the freedom and rights of athletes. Once drafted, the athletes had to be retained in the club until retirement or until the athlete was traded or released. It was until year of 2000 when Free Agency began. Professional baseball players are now eligible to become a free agent after nine years of contract. Free agent players earn rights to make their own decision and negotiate terms and salary. The popularity, recognition, character, and most importantly the performance are taken account to determine a salary. Nevertheless, it is important that only after becoming a free agent they are being judged fairly in terms of contracts and terms.

Table 1. The KBO attendance during regular season of 2000-2018

Year	Games	Total Attendance	Average Attendance
2000	532	2,507,549	4,713
2001	532	2,991,064	5,622
2002	532	2,394,570	4,501
2003	532	2,722,801	5,118
2004	532	2,331,978	4,383
2005	504	3,387,843	6,722
2006	504	3,040,254	6032
2007	504	4,104,429	8144
2008	504	5,256,332	10,429
2009	532	5,925,285	11,138
2010	532	5,928,626	11,144
2011	532	6,810,028	12,801
2012	532	7,156,157	13,451
2013	576	6,441,945	11,184
2014	576	6,509,915	11,302
2015	720	7,360,530	10,223
2016	720	8,339,577	11,583
2017	720	8,400,688	11,668
2018	720	8,073,742	11,214

The Model

Scully (1974) used 1968 and 1969 data of MLB to estimate the marginal revenue product of a professional player by estimating the impact of team-level player performance on team winning and also the impact of team winning on team revenue. According to Scully (1974), it is necessary to make an assumption that gross revenue for the team is related to team's performance, so are the player's performance. Fans attend games to see the team they cheer to win, not to witness individual player's skills. Otherwise, we would underestimate marginal revenue products of those skilled players.

We have approached to calculate the MRP in same manner as what Scully's modeled, but we have changed variables accordingly. Not to mention the league itself is different, but also there are huge periodical, cultural, and systemic gap towards analyzing the baseball period.

1. The Team winning percentage model

We observe team i = 1,2,...,N level data at the t = 1,2,...,T regular season. The error term is assumed to be independent and identically distributed (i.i.d.).

$$PCTWIN_t^i = \beta_0 + \beta_1 T.SA_t^i + \beta_2 T.SW_t^i + \beta_3 CONT_t^i + \beta_4 OUT20_t^i + \beta_5 \mathrm{EXP}_t^i + \varepsilon_t^i \quad (1)$$

The model Eq.(1) is the exactly the same as Scully (1974) model. The dependent variable PCTWIN is the team-level winning percentage. It is adjusted as games won divided by total games times 1000. TSA is the team-level slugging average (SA) which is a total bases divided by total at bats 1000. TSW is the team-level strikeout to walk ratio (K/BB) expressed as strikeouts divided by walks multiplied by 100. CONT is a dummy variable which indicates 1 for pennant winners and their rivals at the end of the season, 0 for the team with five or fewer games out of the placing. OUT20 is a dummy variable that assign 1 for teams at the end of the season were twenty or more games out of the placing and 0 otherwise. EXP dummy variable, is equal to 1 if the team is an expansion club and 0 otherwise.

$$PCTWIN_t^i = \beta_0 + \beta_1 T.OPS_t^i + \beta_2 T.WHIP_t^i + \beta_3 CONT_t^i + \beta_4 OUT20_t^i + \beta_5 \mathrm{EXP}_t^i + \varepsilon_t^i \quad (2)$$

In line with And Nam et al. (2012) and Sun (2019), the two performance variables, *TOPS* and *TWHIP* are included in the model Eq.(2) instead of *TSA* and *TSW* for a purpose of robustness check. *TOPS* is the team-level on base plus slugging (OPS) as performance indicator of hitter. It is a sum of a player's on-base percentage and slugging percentage multiplied by 1000. *TWHIP* is the team-level walks plus hits per inning pitched (WHIP), reflecting pitcher's performance. It is the sum of a pitcher's walks and hits, divided by his total innings pitched multiplied by 100.

$$PCTWIN_t^i = \beta_0 + \beta_1 T.WRCplus_t^i + \beta_2 T.ERAplus_t^i + \beta_3 CONT_t^i + \beta_4 OUT20_t^i + \beta_5 \mathrm{EXP}_t^i + \varepsilon_t^i \quad (3)$$

For model Eq.(3), we use *TWRCplus* and *TERAplus* as a performance indicator in order to take account of external factors such as setups for a ballpark and the type of a league. These two variables can be obtained by solving complicated formula which is explained in Table 2. *TWRCplus* is a weighted runs created plus (wRC+). The sign + indicate control of park effect. League average is set for 100, and every point above or below 100 is a percentage point above or below league average. *TERAplus*, earned run average plus (ERA+), is normalized ERA across the entire league, accounting for external factors like ballparks and the matched contender. The average ERA+ is also set to be 100, and a score above or below 100 indicates that the pitcher performed better or worse than the league average.

2. The Team Revenue Model

We illustrate a team revenue model. We observe team level data at the regular season. The error term is assumed to be independent and identically distributed (i.i.d.).

$$REVENUE_t^i = \alpha_0 + \alpha_1 PCTWIN_t^i + \alpha_2 CITYPOP_t^i + B_3 STAD_t^i + B_4 MGAME_t^i + B_5 CHAMP_t^i + B_5 FAN_t^i + \varepsilon_t^i \quad (4)$$

The team revenue model Eq.(4) includes different variables of what originally Scully (1974) proposed. The dependent variable *REVENUE* is just a total of entrance fee collected during the regular season. Other possible source of revenue that were included in Scully (1974)'s work is omitted due to inconsistent availability and limited data. The first and the principal independent variable *PCTWIN* is the team-level winning percentage. It is adjusted as games won divided by total games times 1000. *CITYPOP* is the total population of the city the team based in. *STAD* is seating capacity of stadium. *MGAME* is the number of matches that a team have for a regular season. *CHAMP* is a dummy variable indicates 1 for a winner of a regular season and 0 for otherwise. *FAN* is a dummy variable adjusts intensity of fan interest among the teams.

DATA

1. The Team Winning Percentage Model

In order to drive winning percentage model, the regular season records from 2000 to 2018 was collected from KBO official baseball guide and STATIZ website (www.statiz.co.kr). The total of 162 observations and three following equations include diverse performance indicators. For the Eq.(1), team level winning percentage (*PCTWIN*), team slugging average (*TSA*), team strike out to walk (*TSW*) was used. For the Eq.(2), TSA and TSW are replaced by additional explanatory variables for robustness check. It is team level on base plus slugging average (*TOPS*) and team level walks plus hits per inning pitched (*TWHIP*). The Eq.(3) uses weighted run created (*TWRCplus*) and earned run average plus (*TERAplus*) instead. Dummy variables *CONT, OUT20*, and *EXP* are used in all equations. How all variables are defined and modified is explained in Table 2.

Table 3 presents the descriptive statistic of the KBO records from 2000 to 2018. *PCTWIN* is 0.499 with the maximum value of 695 and the minimum with 265. *TSA* which indicates how many bases a hitter can advance per bat had the average value of 0.409 and also means team slugging average of 408.95. *TSW* measures the ability of pitchers has the average of 183.94 and it implies team level walk out ratio of 1.84.

TOPS explains hitter's capability of reach base and how well a player can hit. It is widely accepted hitter's performance indicator. The average of *TOPS* in the data is 757.76 which is converted from 0.758 to be adjusted to the model. *TWHIP* which counts pitcher's error to let batters to reach base has the average of *TWHIP* is 144.44

Table 2. Variable Definition: The Team Winning Percentage Model

Variable	Description
PCTWIN	dependent variable; team-level win percentage, games won divided by total games times 1000
TSA	team-level hitter performance; slugging average (SA) expressed as total bases divided by total at bats times 1000
TSW	team-level pitcher performance; strikeout to walk (K/BB) ratio expressed as strikeouts divided by walks times 100
TOPS	team-level hitter performance; on base plus slugging (OPS) expressed as adding on-base percentage (OBP) and slugging percentage (SLG) to get one number that unites the two times 1000
TWHIP	team-level pitcher performance; walks plus hits per inning pitched (WHIP) expressed as the number of base runners a pitcher has allowed per inning times 100
TWRCplus	team-level hitter performance; team weighted run created plus calculation formula is as follows: (wRC+) = [{(wRAA / PA + League R / PA) + (League R / PA – Park Factor × League R / PA)} / (AL or NL wRC / PA excluding pitchers)] × 100 where weighted runs above average (wRAA) measures how many runs a hitter contributes, compared with an average player wRAA measures how many runs a hitter contributes, compared with an average player. PA means plate appearance.
TERAplus	team-level pitcher performance; earned run average plus (ERA+) = league ERA, adjusted for park factors × 100 / team ERA where ERA = earned runs allowed/innings pitched × game innings
CONT	dummy variable; 1 for pennant winners and their closest competitors at the end of the season, if the competitor was five or fewer games out and 0 otherwise
OUT20	dummy variable; 1 for teams which at the end of the season were twenty or more games out of the placing and 0 otherwise
EXP	dummy variable; 1 if the team is an expansion club and 0 otherwise

Table 3. Descriptive Statistic (the KBO, 2000-2018)

Variable	#obs	Mean	Std Dev	Min	Max
PCTWIN	162	499.09	79.42	265	695
TSA	162	408.95	36.84	350	510
TSW	162	183.94	33.16	118	282
TOPS	162	757.76	48.69	661	891
TWHIP	162	144.44	10.36	124	176
TWRCAplus	162	100.02	9.19	76.70	122.70
TERAplus	162	100.85	10.32	77.50	129.10
CONT	162	0.23	0.42	0	1
OUT20	162	0.40	0.49	0	1
EXP	162	0.02	0.16	0	1

and is team level walks plus hits per inning pitched of 1.444.

TWRCplus, team weighted run created plus, measures the difference between the object and the league average including ball-park factor. The ball-park factor considers structural conditions such as lighting and fence height so that it can consider hitter-friendly field or pitcher-friendly fields. Our data has the average of 100.02 with minimum of 76.7 and maximum of 122.7. *TERAplus,* team earned average plus, is the most commonly accepted statistical tool for evaluating pitchers. It represents the number of earned runs a pitcher allows per nine innings adjusting ball-park factor. Our data indicates average of 100.85 with minimum of 77.5 and maximum of 129.1.

The variable *CONT* is the dummy variable of 1 for pennant winners and their rivals for the season. The average of 0.23 is signifying that 10 out of 2 teams are competitive for league championship with less than five win-counts apart. The *OUT20* assigns 1 for teams with final score twenty or more games behind the lead team. The average of is 0.40 and it signals average of 4 teams are having a bad season. *EXP* puts 1 for new expansion team and has the average of 0.02. There have been only four new tea during 2000 to 2018.

2. The Team Revenue Model

The team revenue model includes demographical data that are collected from websites of KBO and KOSTAT. For Eq.(4), the team revenue (*REVENUE*), team level winning percentage (*PCTWIN*), population of the city (*CITYPOP*), seating capacity of stadium (*STAD*), number of games matched (*MGAME*), champion of regular season (*CHAMP*), intensity of fan interest (*FAN*) is used. How all variables are defined and modified is explained in Table 4.

Table 5 is the descriptive statistic of the team revenue function. *REVENUE* has the average of 3.4 billion won with minimum of 233 million won

Table 4. Variable Definition: The Team Revenue Model

Variable	Description
REVENUE	dependent variable; team revenue collected by tickets sold in regular season unit of measure: ₩10,000
PCTWIN	independent variable; team-level win percentage, games won divided by total games times 1,000
CITYPOP	independent variable; population of the city the club based in is included to adjust for the magnitude of income unit of measure: 1,000
STAD	independent variable; the seating capacity of stadium that the team based in.
MGAME	independent variable; number of games matched in the regular season
CHAMP	dummy variable; 1 if the team is champion of the regular season and 0 otherwise
FAN	dummy variable; designed to capture intensity of fan interest 1 if the team's revenue is greater than the median, number of CONT is less than two, and OUT20 more than ten times, and 0 otherwise.

Table 5. Descriptive Statistic (the KBO, 2000-2018)

Variable	#obs	Mean	Std Dev	Min	Max
REVENUE	162	346492.74	286891.17	23393.33	1067949.18
PCTWIN	162	499.09	79.42	265	695
CITYPOP	162	4516.84	3692.29	948	10139
STAD	162	18997	6975.4	10000	26800
MGAME	162	133.78	6.48	126	144
CHAMP	162	0.12	0.32	0	1
FAN	162	0.35	0.48	0	1

and maximum of 10.7 billion won. Haitai was the lowest revenue in the data, and it was their last year before they defunct. Doosan Bears had the most revenue collected in 2018. The average of *CITYPOP* is 4.5 million people. This figure is relatively high because LG, Doosan, and Kiwoom are all based in Seoul.

The average number of *STAD* is nineteen thousands and more details of each teams stadium can be found in the Table 6. Busan Sajik stadium of Lotte Giants has the biggest seating capacity with almost twenty seven thousands seats. KIA Tigers moved their stadium to Champions Fields in 2014. Kiwoom Heroes moved to first domed baseball stadium located in Seoul in 2016. Samsung had the smallest capacity with only ten thousands seats, but they moved to Lions Park with seating capacity of 24000 in 2016. The mean value of *MGAME* is 133.78. The number of games matched has been changing accordingly with the expansion of teams. SK Wyverns joined in 2000, Kiwoom Heroes joined in 2008, NC Dinos in 2013, and KT Wiz joined in 2015. The mean value for *CHAMP* is 0.12 because it is only the recent that the number of team have become ten from eight. The variable was additionally added to capture the fan's royalty. We assign 1 to Lotte, LG, and Hanwha. The reason is primarily the amount of revenue and the team performance. Our data in Table 6 indicates that LG had a good season only once in 2013, and Lotte, and Hanwha never were the winner or the contender throughout the whole regular season since 2000. However, they were included more than 10 times for a bad performance indicator. Despite the assumption that fans cheer for their home team to win, these three teams were very poor with the performance but numbers were high in tickets selling. The only plausible explanation would be the intensity of fan interest. Also, the variable *FAN* differentiate Doosan and LG as they share the same host city and the stadium.

Table 6. Team Information (the KBO, 2000-2018)

Team	Number of CONT	Number of OUT20	Host city	Stadium Capacity
KIA Tigers	3	9	Gwangju	12,500 ↓ 22,244
Lotte Giants	0	10	Busan	26,800
LG Twins	1	12	Seoul	25,553
Doosan Bears	8	2	Seoul	25,553
Samsung Lions	12	3	Daegu	10,000 ↓ 24,000
Hanwha Eagles	0	12	Daejeon	13,000
SK Wyverns	6	1	Incheon	26,000
Kiwoom Heroes	2	4	Seoul	10,500 ↓ 16,731
NC Dinos	1	1	Changwon	11,000
KT Wiz	0	2	Suwon	22,067

Estimation Results

1. The Team Winning Percentage Model

The estimation results of the team winning percentage model are described in the Table 7. The correlation coefficients between the variables are measured and compared. As there is no indubitable test of whether multicollinearity is a problem, a robustness test is conducted. Dropping a variable – either *CONT* or *OUT20* – does not qualitatively change the results across model specifications. Full results are available upon request. Taking a look at normal plot of the residuals via the STATA *pnorm* command, we can accept that the residuals are reasonably close to normal. The results show a trivial deviation from normality. The same result applies to the team revenue model diagnostics. Team dummy variables are included in specifications (2), (4), and (6). All of the coefficients are statistically significant at the 1% level with expected signs except EXP.

The coefficients for *TSA* and *TSW* under specification (1) using Eq.(1) are 0.34 and 0.32, respectively. This means that each one point increase in team slugging average, the hitter indicator, will raise the team winning percentage by 0.34%. This applies for team strike to walk ratio as well. One point increase in *TSW* will attribute to the team winning percentage by 0.32. They are statistically significant at the 1% level and its coefficient is logically acceptable.

The point estimate robustness check requires comparisons of the estimate results among the three sets of player performance measures. The estimation results under specifications (3)-(4) using Eq.(2) suggest that *TOP* has a positive sign of 0.67 and it tells a one-hundredth point increase in this batting indicator will raise the team winning percentage by 0.67. *TWHIP* has a

Table 7. Estimation Results for the Team Winning Percentage Model (2000-2018)

Variable	(1)		(2)		(3)		(4)		(5)		(6)	
TSA	0.342***	(0.09)	0.375***	(0.09)								
TSW	0.318***	(0.11)	0.355***	(0.11)								
TOPS					0.673***	(0.07)	0.687***	(0.08)				
TWHIP					-2.949***	(0.28)	-2.864***	(0.31)				
TWRCplus									3.191***	(0.36)	3.182***	(0.35)
TERAplus									3.481***	(0.26)	3.518***	(0.27)
CONT	69.88***	(7.08)	65.10***	(8.20)	48.04***	(6.30)	46.18***	(7.40)	27.25***	(5.91)	23.77***	(6.50)
OUT20	-83.27***	(6.99)	-78.73***	(6.92)	-54.67***	(5.68)	-55.48***	(5.83)	-35.35***	(4.99)	-35.25***	(5.10)
EXP	-38.52**	(16.68)	-23.31	(18.10)	-22.86***	(6.40)	-13.21	(8.93)	-27.96***	(7.30)	-21.05***	(7.76)
Constant	318.7***	(34.43)	309.7***	(33.96)	426.3***	(42.76)	416.0***	(42.23)	-162.7***	(49.29)	-154.4***	(49.98)
No. of observations	162		162		162		162		162		162	
$AdjR^2$	0.783		0.794		0.856		0.856		0.904		0.907	
Equation	(1)		(1)		(2)		(2)		(3)		(3)	
Team id dummy variables	X		O		X		O		X		O	

Dependent variable: *PCTWIN*
Standard errors in parentheses $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

negative sign of -2.95 which logically makes sense that the winning percentage decrease as a number of runners a pitcher has permitted per inning increases.

The coefficients for *TWRCplus* and *TERAplus* under specification (5) using Eq.(3) are 3.19 and 3.48, respectively. The coefficient indicates that one point increase in *TWRCplus* and *TERAplus* would have sufficient impact the team winning percentage.

All of the coefficients of the dummy variables, *CONT, OUT20,* and *EXP* have the expected sign. The estimated impacts of *CONT* on team winning percentages are positive. The estimated coefficients for *OUT20* are negative, capturing the opposite aspect of *CONT.* The negative coefficients of *EXP* explain that newly contracted team would finish the season relatively worse than the average. The estimates for *EXP* are of statistical significant at 5 to 10%. This relatively low level of significance could be because there are only three observations throughout the tested periods.

2. The Team Revenue Model

Table 8 shows the team revenue estimation results from 2010 to 2018. Review of the pairwise correlation coefficients using data did not reveal any highly correlated variables. Thus, we proceed with the chosen variables. *PCTWIN*, *CITYPOP*, *STAD*, and *MGAME* are all significant at the 1% level with expected signs. The estimation result shows that one-point increases in *PCTWIN* yields 648.81 (₩6,488,100) worth of revenue. One-point increase in *CITYPOP*, which is a thousand people, raise revenue as 24.4 worth of revenue. One-point increase in *STAD* and *MGAME* will result 8.52 and 23058.88 worth of revenue. We can recognize that variables directly related to operations of baseball game have larger coefficient. The result shows that variable *CHAMP* is not statistically significant. The estimated coefficient is 23752.13 which is the opposite effect of common belief. The regular season champion negatively affects the team revenue is unreliable result. The variable *FAN* is statistically significant at 10% level and has the estimated coefficient of 69164.01. The number indicates that team with intense fan interest would have more revenue.

Table 8. Estimation Results for the Team Revenue Model (2000-2018)

	(1)	
PCTWIN	648.81***	(242.70)
CITYPOP	24.44***	(5.12)
STAD	8.52***	(2.34)
MGAME	23058.88***	(2066.40)
CHAMP	-23752.13	(48860.10)
FAN	69164.01*	(40926.10)
Constant	-3355917.8***	(300475.60)
No. of observations	162	
$AdjR^2$	0.493	
Equation	(1)	

Dependent variable: *REVENUE*
Standard errors in parentheses *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$

3. Marginal Revenue Product Estimates

The MRP estimates of hitters and pitchers are calculated with use of estimates for the team winning percentage model Eqs.(1)-(2) and the team revenue model Eq.(4) in Table 9. The individual player' MRP is derived from the assumption that team performance is simply the linear summation of individual performance following Scully (1974). We divided TSA with the number of team bats (TAB) of a season, and multiplied it with actual records of bats (AB) to get an individual hitter performance.

To get an individual pitcher performance, we divided TSW with team total inning of a season (T-inning) and multiplied with an individual's total inning of a season.

From Eq.(1), one point increase in TSA (TSW) increases by 0.34 (0.32). From Eq.(4), empirical results indicate that raising the team-level win percentage one point increase team revenue by ₩6,488,100. Therefore, the MRP of hitters and pitchers is derived as

MRP hitters = 0.34 × ₩6,488,100
= ₩2,205,954 per point TSA
MRP pitchers = 0.32 × ₩6,488,100
= ₩2,076,192 per 1/100 point TSW

The team revenue function, however, had its limitation with including sources of revenue except ticket sales. Income from advertisement, broadcasting, and goods selling are omitted from the team revenue function due to unavailability and inconsistency of data. The ratios of ticket sales to total revenue range from 5-15% across teams during 2000-2018. Thus, we proceed with chosen ratio of ticket sales to revenue, 10%, for re-calculating the adjusted MRP estimates.

We compare actual stats of FA hitter Lee Dae-ho with the estimated MRPs in Table 9. Lee Dae-ho signed a four-year free agent contract with Lotte Giants in 2017. Specifically, the estimated MRP of Lee Dae-ho in 2017 using winning percent Eq.(1) and revenue Eq.(4) is $₩2,205,954 \times 533 \times \frac{540}{4944} = ₩127,136,099$. Using the Eq.(2) and Eq.(4), it is calculated as $₩2,205,954 \times 925 \times \frac{540}{4944} = ₩220,639,572$. His annual salary is 2.5 billion won in 2017, and the estimated MRPs are 127 (220) million won for the estimates derived from Eq.(1) and Eq.(4)(Eq.(2) and Eq.(4)).

As illustrated in Table 9, along with the rise in performance, the estimated annual salary increases. However, there are significant gap between the estimated MRP compare to actual salary before the FA contract. These undervalue of salary is in accordance with the claim of Scully (1974) that reserve clause cause monopsonistic exploitation of professional baseball players. According to the implied adjusted MRP estimate, Lee Dae-ho's actual annual salary is about 1.1 to 2 times bigger than the estimated values after the FA contract. While it cannot be certain which of these equations is the most accurate, Using Eq.(2) resulted a value closer to the actual salary.

We compare actual stats of FA pitcher Yang Hyun-jong with the estimated MRPs in Table 10. KIA Tiger pitcher Yang Hyun-jong signed a one-year deal in 2017. The estimated MRP of Yang in 2017 using Eq.(1) and Eq.(4) is $₩2,076,192 \times 351 \times \frac{193.10}{1319} = ₩106,687,149$T h e adjusted MRP estimate is $₩1,066,871,490$. His estimated MRP for the year 2017 is 106 million won and his actual salary is 1.5 billion won. The actual salary is 1.5 times bigger than the adjusted MRP estimate. Again, there are some possible causes of difference that we could not take account of, but we are able to see the same pattern for both examples. According to the estimated results, the both seemed to be underpaid. before the FA contract and are now overpaid after the FA contract.

Table 9. Stats of Lee Dae-ho: Estimate MRP and Actual Salary (Unit: Won)

Year	SA in Eq.(1)	OPS in Eq.(2)	Estimated MRP using Eqs.(1)&(4)	Estimated MRP using Eqs.(2)&(4)	Adjusted MRP using Eqs.(1)&(4)	Adjusted MRP using Eqs.(2)&(4)	Actual salary
2001	500	1.056	1,951,739	4,122,074	19,517,390	41,220,740	-
2002	447	0.792	55,728,206	98,739,909	557,282,060	987,399,090	20,000,000
2003	362	0.689	27,184,862	51,741,355	271,848,620	517,413,550	32,000,000
2004	441	0.772	96,802,917	169,459,982	968,029,170	1,694,599,820	37,000,000
2005	452	0.805	106,703,320	190,035,781	1,067,033,200	1,900,357,810	70,000,000
2006	571	0.98	134,880,996	231,494,529	1,348,809,960	2,314,945,290	130,000,000
2007	600	1.053	128,547,284	225,600,484	1,285,472,840	2,256,004,840	320,000,000
2008	478	0.879	107,773,499	198,185,996	1,077,734,990	1,981,859,960	360,000,000
2009	531	0.908	126,676,658	216,614,701	1,266,766,580	2,166,147,010	360,000,000
2010	667	1.111	150,506,203	250,693,241	1,505,062,030	2,506,932,410	390,000,000
2011	578	1.011	136,680,890	239,073,322	1,366,808,900	2,390,733,220	630,000,000
2017	533	0.925	127,136,099	220,639,572	1,271,360,990	2,206,395,720	2,500,000,000
2018	593	0.987	138,462,959	230,460,271	1,384,629,590	2,304,602,710	2,500,000,000
2019	435	0.79	94,612,958	171,825,833	946,129,580	1,718,258,330	2,500,000,000
2020	452	0.806	109,000,289	194,367,772	1,090,002,890	1,943,677,720	2,500,000,000

Note. Signing bonus excluded

Table 10. Stats of Yang Hyun-jong: Estimate MRP and Actual Salary (Unit: Won)

Year	T-inning	WHIP	SW in Eq.(1)	Estimated value using Eq.(1)&(4)	Adjusted MRP using Eqs.(1)&(4)	Actual salary
2007	1109.3	1.41	155	14,273,001	142,730,010	-
2008	1124	1.65	117	16,251,929	162,519,290	24,000,000
2009	1199	1.29	240	61,589,656	615,896,560	35,000,000
2010	1183.7	1.58	148	43,896,631	438,966,310	100,000,000
2011	1163.7	1.74	107	20,254,692	202,546,920	170,000,000
2012	1183.7	2	84	6,033,533	60,335,329	140,000,000
2013	1142.3	1.36	221	41,854,998	418,549,980	90,000,000
2014	1121.7	1.4	214	67,772,667	677,726,670	120,000,000
2015	1273	1.24	201	60,351,623	603,516,230	750,000,000
2016	1276	1.34	189	61,535,500	615,355,000	750,000,000
2017	1319	1.31	351	106,687,149	1,066,871,490	1,500,000,000
2018	1270.7	1.31	353	106,182,508	1,061,825,080	2,300,000,000
2019	1270.7	1.07	493	148,455,652	1,484,556,520	2,300,000,000
2020	1236.3	1.42	232	67,052,118	670,521,180	2,300,000,000

Conclusion

This study compared the estimated marginal revenue productivity of Korean Professional Baseball League players and the actual annual salary of free agent players based on the wage determination theory of labor economics and multiple regression analysis on panel data of the KBO records from 2000 to 2018. Given the assumption that a team maximizes profits by selecting a level of player inputs and non-player inputs in Scully(1974), the estimates for MRP of a KBO player are derived from team-level performance, team winning, and team revenue. For robustness purposes, recent performance measures of sabermetrics based on different types of batting and pitching records are used and compared.

We limited MRP evaluation to FA players only, avoiding calculation of underpaid values under the reserve clause. Using the estimated coefficients of team winning percentage model and team revenue model equations, MRP estimate for individual free agent player is calculated using each player's stats. MRP estimates of Lee Dae-ho and Yang Hyun-jong, arguably the best player for their position as a hitter and a pitcher, are compared with their actual salaries. Both players signed their first FA contract in Korea in 2017. In line with Sommers and Quinton (1982), the empirical findings using Korean professional baseball players also confirm that both players are underpaid in pre-FA period and are overpaid in post-FA period. These gaps between the estimated MRP and actual salary can also be attributed to the limited access to sources of team revenue data. Evaluations based on the adjusted MRP estimate resulted a value closer to the actual salary.

However, we cannot simply conclude that Lee Dae-ho and Yang Hyun-Jong are being overpaid, as the estimation does not take account of adjustment of the collective ratio of ticket sales and players' salary negotiation ability. A potential problem in evaluating the MRP estimate may also arise from an additive functional form assumption in the team revenue equation. Also, the variable *FAN*, the numerical representation of the fan spirit have plenty more room to be developed.

Studies on predicting outcome of a baseball match have been continuously conducted with the beginning of professional baseball history. Yet there is no acknowledged function or mathematical representation that captures all the variables and possibilities of a single baseball game. While our study has not been possible to provide definite answer to the questions, we have converted stats into statistical value of performance to financial value. The analysis of more flexible functional form of the team revenue model is left for future work.

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국문초록

한국프로야구 선수들의 연봉 적절성
: 자유계약 선수를 중심으로

박성준(단국대학교) · 선주연(단국대학교)

본 연구는 노동경제학의 임금결정이론을 기반으로 한국프로야구 자유계약선수들의 기록을 바탕으로 추정된 한계수입생산성과 실제 연봉을 비교하였다. 2000년부터 2018년까지의 패널자료를 구축하여 개별 선수의 한계수입생산성을 실증적으로 분석하였다. 먼저 Scully(1974)의 모형을 한국프로야구에 적용하여 팀 승률 방정식과 팀 수익 방정식을 식별하였다. 다중회귀분석을 통해 타자와 투수 성적지표가 팀 승리에 미치는 영향력을 추정하고, 팀 승리가 총 수익에 미치는 영향력을 추정하였다. 최신 세이버메트릭스 지표를 설명변수로 사용하여 모형의 강건성을 검토하였다. 팀 승률 회귀모형식과 팀 수익 회귀모형식 추정치로부터 개별 선수의 한계수입생산성을 도출하였으며, 이대호 선수와 양현종 선수의 연도별 성적을 이용하여 조정된 연봉 추정치와 실제 연봉을 비교하였다.

※ 주요어 : 한국프로야구, 다중회귀분석, 한계수입생산성, 연봉추정, 세이버메트릭스

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